

Business Analytics with Statistics and Excel

CONTINUOUS ASSESSMENT - CA2 SET B

HR Analytics: Performance, Compensation and Workforce

Duration: 50 minutes	Maximum Marks: 50
Assessment: In-Class Excel Assessment	Submission: Registration No_CA2_SET_B.xlsx

Student Name: _____

Registration No.: _____

Supplied workbook: LPU_QTTM510_CA2_Set_B_HR_Dataset.xlsx

Instructions

- Work only in the supplied workbook. Do not delete or rename the HR_Data, Answer_Sheet, Analysis or Data_Dictionary sheets.
- Use Power Query only where requested and Excel formulas/functions where requested.
- Keep calculations visible and label outputs Q1 to Q6 in the Analysis sheet.
- Record final findings and evidence locations in the Answer_Sheet.
- For statistical questions, show the test statistic, p-value, decision and a one-sentence HR conclusion.
- The dataset is synthetic teaching data and must not be presented as actual company data.

Case context

You are a junior business analyst reviewing 300 employee records. HR management needs a short, evidence-based view of data quality, employee compensation, leave and performance. Complete only the tasks below; no additional analysis is required.

Q1. Power Query: Essential Data Check

8 marks | Suggested: 6 min

Load HR_Data into Power Query and complete these checks:

- Set Join_Date and Last_Promotion_Date as Date. Set Experience_Years, Salary, Performance_Rating, Leave_Days, Job_Satisfaction and the pre/post training fields as appropriate numeric types.
- Check Employee_ID for duplicates and check the table for blank/null values.
- Load one cleaned table for analysis.

Evidence: final row count, duplicate/null result, and Applied Steps.

Q2. Power Query: Two Transformations

8 marks | Suggested: 6 min

Using the cleaned query, create:

- Experience_Band using a Conditional Column: less than 3 years = Fresher/Junior; 3 to 7 years inclusive = Experienced; more than 7 years = Highly Experienced.
- Employee_Code using Column from Examples by extracting the numeric part of Employee_ID. Example: EMP1125 becomes 1125.

Evidence: show both completed columns.

Q3. Excel Functions and Monthly Summary

8 marks | Suggested: 7 min

In Excel, create:

- Join_Month using a date function to display the month name.
- Employee_Code_Excel using a text function to extract the numeric portion of Employee_ID.
- A PivotTable by Join_Month showing Count of Employee_ID and Average of Performance_Rating.

Evidence: show both formulas and the completed PivotTable.

Q4. Two-Tailed One-Sample t-Test

10 marks | Suggested: 10 min

Test whether the mean annual Salary differs from Rs. 60,000 at the 5% significance level. Use the sample standard deviation.

- State H_0 and H_1 .
- Calculate n , sample mean, sample standard deviation, t -statistic and two-tailed p -value.
- State the decision and one-sentence HR conclusion.

Q5. One-Tailed One-Sample z-Test

8 marks | Suggested: 8 min

Assume the population standard deviation of Leave_Days is 4 days. Test whether the population mean is greater than 12 days at the 5% significance level.

- State H_0 and H_1 .
- Calculate the z -statistic and upper-tail p -value.
- State the decision and one-sentence HR conclusion.

Q6. Confidence Interval and Performance Comparison

8 marks | Suggested: 8 min

Complete both parts:

- Calculate a t -based 95% confidence interval for the population mean of Job_Satisfaction. Show the sample mean, lower limit and upper limit.
- Calculate the averages of Pre_Training_Performance and Post_Training_Performance. State whether average performance appears to have improved. This is descriptive only; no additional hypothesis test is required.

Evaluation grid

Question	Skill assessed	Marks
Q1	Essential Power Query data check	8
Q2	Conditional Column and Column from Examples	8
Q3	Text/date functions and PivotTable summary	8
Q4	Two-tailed one-sample t -test	10
Q5	One-tailed one-sample z -test	8
Q6	Confidence interval and descriptive comparison	8
TOTAL		50

Scope: Power Query, Excel text/date functions, PivotTables, hypothesis-testing fundamentals, t -test, z -test and confidence intervals. No regression, ANOVA, forecasting, VBA or advanced modelling is required.